Homework 6

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Problem 1

Let $\pi_1(X) \cong 0$, and let $f : S^1 \to X$ be continuous. WTS $\exists f^* : D^2 \to X$ s.t. $f^*(S^1) = f(S^1)$.

Problem 2

Problem 3

The first two parts are just proving functorality of $\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$. First, we show homotopy is preserved by continuity. Let ϕ, ψ be homotopic paths in X, and let $f : X \to Y$ be a base-preserving continuous map. Let $\Phi : [0,1] \times [0,1] \to X$ be the homotopy $\phi \to \psi$, meaning $\phi = \Phi(0, -)$ and $\psi = \Phi(1, -)$. We thus have $f \circ \phi = f \circ \Phi(0, -)$ and $f \circ \psi = f \circ \Phi(1, -)$, and thus $f \circ \Phi$ clearly induces a homotopy of $f \circ \phi$ and $f \circ \psi$, since composition of continuous maps are continuous. Therefore, homotopy is invariant under continuity.

Note that $\pi_1(f) : \pi_1(X) \to \pi_1(Y)$ will be the map $\pi_1(f)[\phi] = [f \circ \phi]$. We want to show that $\pi_1(f \circ g) = \pi_1(f) \circ \pi_1(g)$. We have

$$\pi_1(f) \circ \pi_1(g)[\phi] = \pi_1(f)[g \circ \phi] = [f \circ g \circ \phi] = \pi_1(f \circ g)[\phi].$$

Therefore, π_1 preserves morphism composition.

[identity proof]

Since π_1 is a functor, if $f : X \to Y$ is a homeomorphism, then it has a two-sided continuous inverse $f^{-1} : Y \to X$. We thus have

$$\pi_1(f) \circ \pi_1(f^{-1}) = \pi_1(f \circ f^{-1}) = \pi_1(\mathrm{id}_Y) = \mathrm{id}_{\pi_1(Y)}.$$

The same logic shows $\pi_1(f)$ and $\pi_1(f^{-1})$ are inverses in **Grp**. Therefore, $\pi_1(X) \cong \pi_1(Y)$.

Problem 4

The idea is going to be that $\pi_1(S^1) = \mathbb{Z}$, but more precisely every time the path fully winds around S^n, it changes. Or at the very least if the map is not surjective, it cannot induce any maps (paths) that go all the way around, and thus any path is null-homotopic.

Let $f : X \to S^n$ be continuous, but not surjective. Let $a \in \operatorname{im} f$. We want to show there exists a homotopy $\Phi : f \to a$ (viewing as a 2-morphism in **Top**).

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I could do this the normal way, but where's the fun in that? Consider the category $\mathbf{Top}_*$ of pointed topological spaces, where morphisms are continuous maps preserving the base point. We need to show we can define the homotopic category $\mathbf{Toph}_*$ as the quotient category $\mathbf{Top}_*/\sim$, where $\sim$ is homtopy. That is, we need to show there exists horizontal composition of homotopy.

I omit mention of the base points for now to avoid notational clutter. It is not difficult to show all definitions and arguments I make are invariant under choice of base points, so any mention of such for now is immaterial. Let $f_1, g_1 : X \to Y$ and $f_2, g_2 : Y \to Z$ be morphisms in $\mathbf{Top}_*$, and let $\Phi : f_1 \Rightarrow g_1$, $\Psi : f_2 \Rightarrow g_2$ be homotopies. We want to show there exists a homotopy $\Psi\Phi : f_2 \circ f_1 \Rightarrow g_2 \circ g_1$. Define $\Psi\Phi : [0,1] \times X \to Z$ as the map $\Psi\Phi(x, y) \coloneqq \Psi(x, \Phi(x, y))$. For now, it suffices to show this is a homotopy. Indeed,

$$\Psi(0, \Phi(0, -)) = \Phi(0, -) \circ \Phi(0, -) = f_2 \circ f_1,$$

and similarly for g. Moreover, this is clearly continuous, as this can be viewed as the induced morphism by the universal property of products:

$$\begin{tikzcd}
 & {[0,1] \times X} \arrow[dl, "\mathrm{id}"'] \arrow[d, dashed, "\zeta"] \arrow[dr, "\Phi"] & \\
{[0,1]} & {[0,1] \times Y} \arrow[l] \arrow[r] \arrow[d, "\Psi"] & Y \\
 & Z &
\end{tikzcd}$$

We can equivalently view $\Psi\Phi$ as $\Psi \circ \zeta$. The fact that ζ is continuous follows from something I proved quite a few homeworks ago and have been using since, and commutivity of the diagram is an easy consequence of this. Since this is continuous, and satisfies $\Psi\Phi(0, -) = f_2 \circ f_1$ and $\Psi\Phi(1, -) = g_2 \circ g_1$, this is a homotopy $f_2 \circ f_1 \Rightarrow g_2 \circ g_1$. Therefore, morphism composition respects homotopy equivalence, and we can define the quotient category $\mathbf{Toph}_* \coloneqq \mathbf{Top}_*/\sim$. That is, $\mathrm{Hom}_{\mathbf{Toph}_*}(X, Y) = \mathrm{Hom}_{\mathbf{Top}_*}(X, Y)/\sim$.

Now we show that the assignment π_1 factors through $\mathbf{Toph}_*$ in such a way that the diagram

$$\begin{tikzcd}
\mathbf{Top}_* \arrow[r, "\pi_1"] \arrow[d, "\pi"'] & \mathbf{Grp} \\
\mathbf{Toph}_* \arrow[ur, "\mathrm{Hom}(S^1, -)"'] &
\end{tikzcd}$$

commutes up to isomorphism. It thus suffices to show π and $\mathrm{Hom}(S^1, -)$ are functors, and that $\pi_1 \cong \pi \circ \mathrm{Hom}(S^1, -)$, since composites of functors are functors. The functor π is the natural projection onto the quotient group, and is defined by $X \mapsto X$ and $f \mapsto [f]_\sim$, for X any topological space, f any morphism, and $\sim$ the homotopy relation.

To show π is a functor, we first have to actually consider the definition of morphisms in the homotopic category. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms in $\mathbf{Top}_*$. We define the composition $[g]_\sim \circ [f]_\sim$ in $\mathbf{Toph}_*$ as $[g \circ f]_\sim$. This is well-defined, since we already showed horizontal composition of homotopies is well-defined up to homotopy; that is, if $f \sim \phi$ and $g \sim \psi$ is given by the diagram

$$\begin{tikzcd} X \arrow[r, bend left, "f"] \arrow[r, bend right, "\phi"'] \arrow[r, phantom, "\Downarrow\mu"] & Y \arrow[r, bend left, "g"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\Downarrow\nu"] & Z, \end{tikzcd}$$

then $g \circ f \sim \psi \circ \phi$ by horizontal composition:

$$\begin{tikzcd} X \arrow[r, bend left, "g\circ f"] \arrow[r, bend right, "\psi\circ\phi"'] \arrow[r, phantom, "\Downarrow\nu\circ\mu"] & Z. \end{tikzcd}$$

It is then clear by definition of $\mathbf{Toph}_*$ that π is functorial:

$$\pi(g \circ f) = [g \circ f]_\sim = [g]_\sim \circ [f]_\sim = \pi(g) \circ \pi(f),$$

and for any $f : Z \to X$ and $g : X \to Z$,

$$[1_X]_\sim \circ [f]_\sim = [1_X \circ f]_\sim = [f]_\sim,$$

$$[g]_\sim \circ [1_X]_\sim = [g \circ 1_X]_\sim = [g]_\sim$$

since 1_X is an identity in $\mathbf{Top}_*$.

Next, we show that $\mathrm{Hom}_{\mathbf{Toph}_*}(S^1, -) : \mathbf{Toph}_* \to \mathbf{Grp}$ is a functor; that is, we show there is a group structure on $\mathrm{Hom}_{\mathbf{Toph}_*}((S^1, s), (X, x_0))$ for all $(X, x_0) \in \mathbf{Toph}_*$, and that the induced morphisms are group homomorphisms. The structure in question is clear; for any two morphisms $[\phi_1]_\sim, [\phi_2]_\sim : (S^1, s) \to (X, x_0)$, we define the concatenation $[\phi_2]_\sim \cdot [\phi_1]_\sim$ the same as we did for loops. Since S^1 is parametrized by $(\cos\theta, \sin\theta)$, we can write ϕ_2, ϕ_1 as functions of θ, and define

$$\phi_2 \cdot \phi_1 := \begin{cases} \phi_1(2\theta), & 0 \le \theta < \pi, \\ \phi_2(2\theta), & \pi \le \theta < 2\pi \end{cases}.$$

Both $\phi_i(2\theta)$-s are continuous by previous arguments, as they are a composition of ϕ_i and multiplication by 2.

Concatenation is well-defined up to homotopy by the exact same proof as done in class, except substituting S^1 for $[0, 1]$, and thus getting rid of the requirement that $\phi(0) = \phi(1)$, since continuity requires that in the limit as $x \to s$ from either direction, $\phi(x) \to \phi(s)$ from both directions. Moreover, $\mathrm{Hom}_{\mathbf{Toph}_*}((S^1, s), (X, x_0))$ is then manifestly closed under concatenation. Concatenation is also associative up to homotopy, since associativity clearly amounts to a reparametarization of S^1, and as shown in class reparamaterizations are homotopic. The proof we did for $[0, 1]$ generalizes immediately to S^1.

Now we show the homotopy class of the constant map $x \mapsto x_0$ is the identity.

Finally, we show this is the same as π_1. First, we show that $\mathrm{Hom}_{\mathbf{Top}_*}((S^1, s), (X, x_0))$ is precisely the set of loops in X at x_0. Consider the map $\alpha : \mathrm{Loop}(X, x_0) \to \mathrm{Hom}((S^1, s), (X, x_0))$ given by $\phi(x) \mapsto (\cos(2\pi\phi(x)), \sin(2\pi\phi(x))$.

the plan here is going to be to consider the quotient (sets) by $0 \sim 1$, which allows us to inuqely identify loops with morphisms. we can attempt to define an inverse map and show it is well-defined

- Show that every loop factors uniquely through it in such a way that the induced map is continuous;

- Show that the map is invertible;
- Conclude that the set of loops is isomorphic to the set of morphisms.

Finally, we show that this is indeed a group *isomorphism*, concluding that the diagram commutes up to natural isomorphism.

To conclude the proof, we can recover functorality of π_1 from the natural isomorphism.

Lemma 1. *Let $\mathcal{F} : \mathcal{C} \to \mathcal{D}$ be a functor, and let $\mathcal{G} : \mathcal{C} \to \mathcal{D}$ be an assignment* $\mathrm{Obj}(\mathcal{C}) \to \mathrm{Obj}(\mathcal{D})$ *and* $\mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{C}}(\mathcal{G}(A), \mathcal{G}(B))$ *for all $A, B \in \mathcal{C}$. Let $\eta : \mathcal{F} \Rightarrow \mathcal{G}$ be a natural isomorphism, in the sense that for all $A, B \in \mathcal{C}$, and for all $f : A \to B$ in $\mathcal{C}$, the diagram*

$$\begin{tikzcd}
\mathcal{F}(A) \arrow[r, "\mathcal{F}(f)"] \arrow[d, "\eta_A"'] & \mathcal{F}(B) \arrow[d, "\eta_B"] \\
\mathcal{G}(A) \arrow[r, "\mathcal{G}(f)"'] & \mathcal{G}(B)
\end{tikzcd}$$

commutes, and each η_A is an isomorphism. Then $\mathcal{G}$ is a functor.

Proof. First, let

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

be morphisms in $\mathcal{C}$. By assumption, the diagram

$$\begin{tikzcd}
\mathcal{F}(X) \arrow[r, "\mathcal{F}(f)"'] \arrow[rr, bend left, "\mathcal{F}(g \circ f)"] \arrow[d, "\eta_X"'] & \mathcal{F}(Y) \arrow[r, "\mathcal{F}(g)"'] \arrow[d, "\eta_Y"'] & Z \arrow[d, "\eta_Z"'] \\
\mathcal{G}(X) \arrow[r, "\mathcal{G}(f)"'] & \mathcal{G}(Y) \arrow[r, "\mathcal{G}(g)"'] & \mathcal{G}(Z)
\end{tikzcd}$$

commutes in $\mathcal{D}$. Additionally,

$$\begin{tikzcd}
\mathcal{F}(X) \arrow[r, "\mathcal{F}(g \circ f)"] \arrow[d, "\eta_X"'] & \mathcal{F}(Z) \arrow[d, "\eta_Z"'] \\
\mathcal{G}(X) \arrow[r, "\mathcal{G}(g \circ f)"'] & \mathcal{G}(Z)
\end{tikzcd}$$

commutes. We thus have

$$\eta_Z \circ \mathcal{F}(g \circ f) \circ = \mathcal{G}(g \circ f) \circ \eta_X.$$

By commutivity of the first diagram,

$$\eta_Z \circ \mathcal{F}(g \circ f) = \eta_Z \circ \mathcal{F}(g) \circ \mathcal{F}(f) = \mathcal{G}(g) \circ \mathcal{G}(f) \circ \eta_X.$$

Therefore,

$$\mathcal{G}(g) \circ \mathcal{G}(f) \circ \eta_X = \mathcal{G}(g \circ f) \circ \eta_X.$$

Since η_X is an isomorphism, there exists an inverse morphism η_X^{-1} such that $\eta_X \circ \eta_X^{-1} = 1_{\mathcal{G}(X)}$, and thus

$$\mathcal{G}(g \circ f) \circ 1_{\mathcal{G}(X)} = \mathcal{G}(g) \circ \mathcal{G}(f) \circ 1_{\mathcal{G}(X)}.$$

Since identity morphisms are identities with respect to composition, we have

$$\mathcal{G}(g \circ f) = \mathcal{G}(g) \circ \mathcal{G}(f).$$

To show identities are preserved, note that the diagram

$$\begin{tikzcd}
\mathcal{F}(X) \arrow[r, "\mathcal{F}(1_X)"] \arrow[d, "\eta_X"'] & \mathcal{F}(X) \arrow[d, "\eta_X"'] \\
\mathcal{G}(X) \arrow[r, "\mathcal{G}(1_X)"'] & \mathcal{G}(X)
\end{tikzcd}$$

commutes. Since $\mathcal{F}$ is a functor, $\mathcal{F}(1_X) = 1_{\mathcal{F}(X)}$, and by commutivity we have

$$\eta_X \circ 1_{\mathcal{F}(X)} = \mathcal{G}(1_X) \circ \eta_X.$$

Since identity morphisms are identities with respect to composition, and since η_X has an inverse, we have

$$\begin{aligned}
&\eta_X = \mathcal{G}(1_X) \circ \eta_X \\
\implies\ &\eta_X \circ \eta_X^{-1} = \mathcal{G}(1_X) \circ \eta_X \circ \eta_X^{-1} \\
\implies\ &1_{\mathcal{G}(X)} = \mathcal{G}(1_X) \circ 1_{\mathcal{G}(X)} \\
\implies\ &1_{\mathcal{G}(X)} = \mathcal{G}(1_X).
\end{aligned}$$

Therefore, $\mathcal{G}$ is a functor. ■